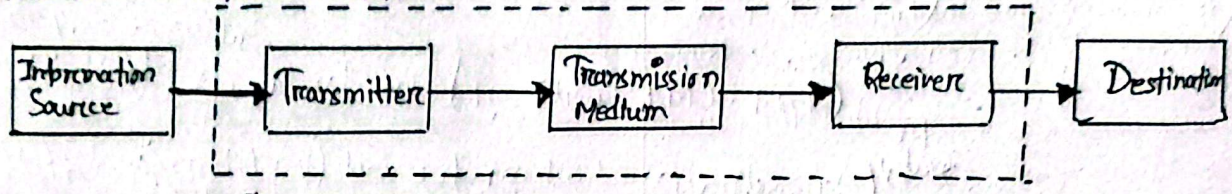


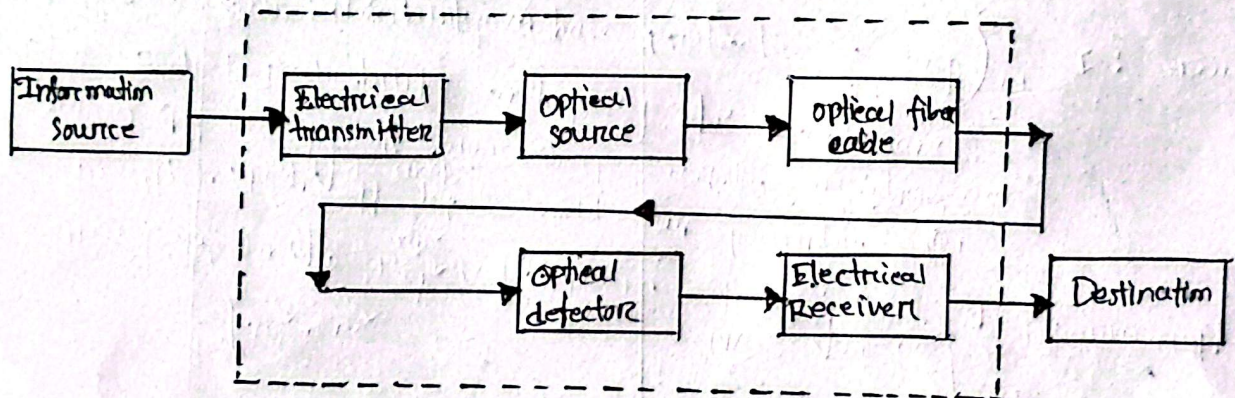
Chapter: 1

Optical Fiber Communication is a method of transmitting information from one place to another in the form of light through an optical fiber.

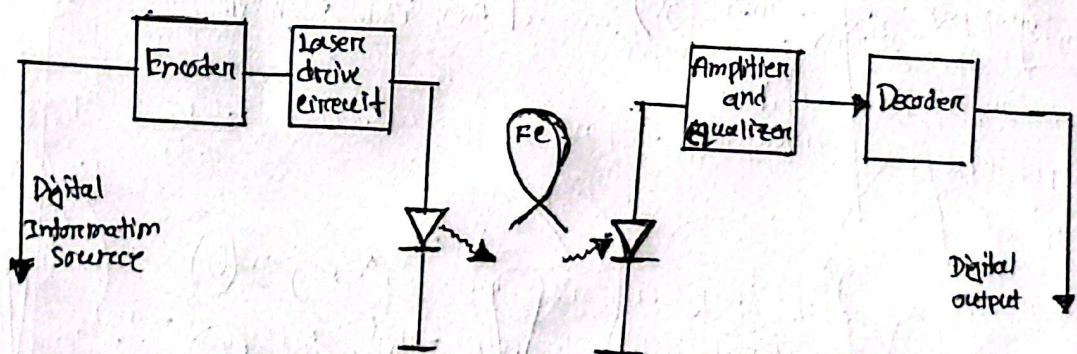
Overall Communication System:



Optical Fiber Communication System:



A digital optical fiber link using Semiconductor laser and an avalanche photodiode detector



Advantage of optical-fiber Communication:

- Enormous potential bandwidth ($10^{13} - 10^{16}$)
- Small size and weight
- Electrical Isolation (do not exhibit interface problem)
- Immunity to interference (free from EMI, RFI)
- Signal security (does not radiate significantly)
- Low transmission loss
- Ruggedness and flexibility (have protective coating)
- System ~~reliability~~ reliability and easy to maintenance
- Potential Low cost.

Chapter: - 2

Snell's Law: $n_1 \sin \theta_1 = n_2 \sin \theta_2$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1}$$

$\therefore$ Critical Angle (θ_c) $\sin \theta_c = \frac{n_2}{n_1}$

$$n_0 \sin \theta_1 = n_1 \sin \theta_2$$

$$\theta = \frac{\pi}{2} - \theta_2$$

$$n_0 \sin \theta_1 = n_1 \sin \left(\frac{\pi}{2} - \theta \right)$$

$$n_0 \sin \theta_1 = n_1 \cos \theta$$

$$n_0 \sin \theta_1 = n_1 \sqrt{1 - \sin^2 \theta}$$

From figure:

$$n_0 \sin \theta_c = n_1 \sqrt{1 - \sin^2 \theta_c}$$

$$= n_1 \sqrt{1 - \frac{n_2^2}{n_1^2}}$$

$$= \sqrt{n_1^2 - n_2^2}$$

NA = numerical aperture is defined as

$$NA = n_0 \sin \theta_c = \sqrt{n_1^2 - n_2^2}$$

for air = 1

$$\therefore \sin \theta_c = \sqrt{n_1^2 - n_2^2}$$

Incident meridional rays over the range $0 \leq \theta_1 \leq \theta_c$ propagated within the fiber

$$k = \frac{n_1^2 - n_2^2}{2n_1^2}$$

$$= \frac{(n_1 - n_2)(n_1 + n_2)}{2n_1^2}$$

$$= \frac{(n_1 - n_2) 2n_1}{2n_1^2}$$

$$= \frac{n_1 - n_2}{n_1}$$

$$NA = \sqrt{n_1^2 - n_2^2}$$

$$\therefore 2 \times n_1^2 = n_1^2 - n_2^2$$

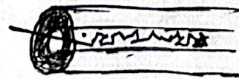
$$\sqrt{2 \times n_1^2} = \sqrt{n_1^2 - n_2^2}$$

$$n_1 \sqrt{2} = \sqrt{n_1^2 - n_2^2}$$

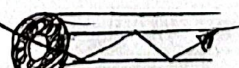
Proved

Types of Rays:

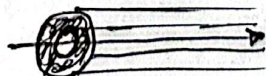
(i) Skew Ray



(ii) Meridional Ray



(iii) Axial Ray



Chapter - 3

Attenuation: The reduction of power of light as it passes through an optical fiber called attenuation. It is measured in decibels per km (dB/km)

Cause of Attenuation:

- Absorption
- Scattering
- Bending loss.

$$\text{Number of decibels (dB)} = 10 \log_{10} \frac{P_i}{P_o}$$

$$\text{The numerical I/O power ratio } \left(\frac{P_i}{P_o} \right) = 10^{\frac{\text{dB}}{10}}$$

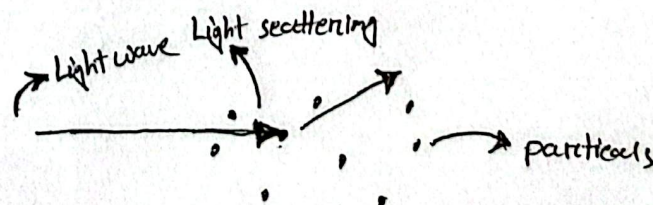
$$\alpha_{\text{dB}} \times L = 10 \log_{10} P_i/P_o$$

where α_{dB} is the signal attenuation per unit length in dB, which is also called loss parameter and L is the fiber length.

Types of Loss:

- Material absorption
 - Intrinsic Absorption: Caused by pure material of the optical fiber itself
 - Extrinsic Absorption: Caused by impurities of the fiber material.
- Material scattering
- Bending loss
- Mode coupling Radiation loss
- Leaky modes loss

Linear scattering loss: occurs when light is scattered in different directions as it travels through the optical fiber. The scattering is called linear because the loss is directly proportional to the input signal power.



Rayleigh scattering: is the scattering of light due to microscopic variation of the refractive index of the fiber causing signal loss

Fiber bend loss: is the loss of optical signal power that occurs when an optical fiber is bent or curved. When the fiber is bent, some of light rays escape from core into cladding and are lost.

Dispersion: is a spreading of a light pulse as it through the fiber. Different parts of light signal travel at different times and different speed, causing the signal spread out.

Effect:

- Pulse overlap
- signal becomes unclear at the receiver
- Limit data transfer speed.

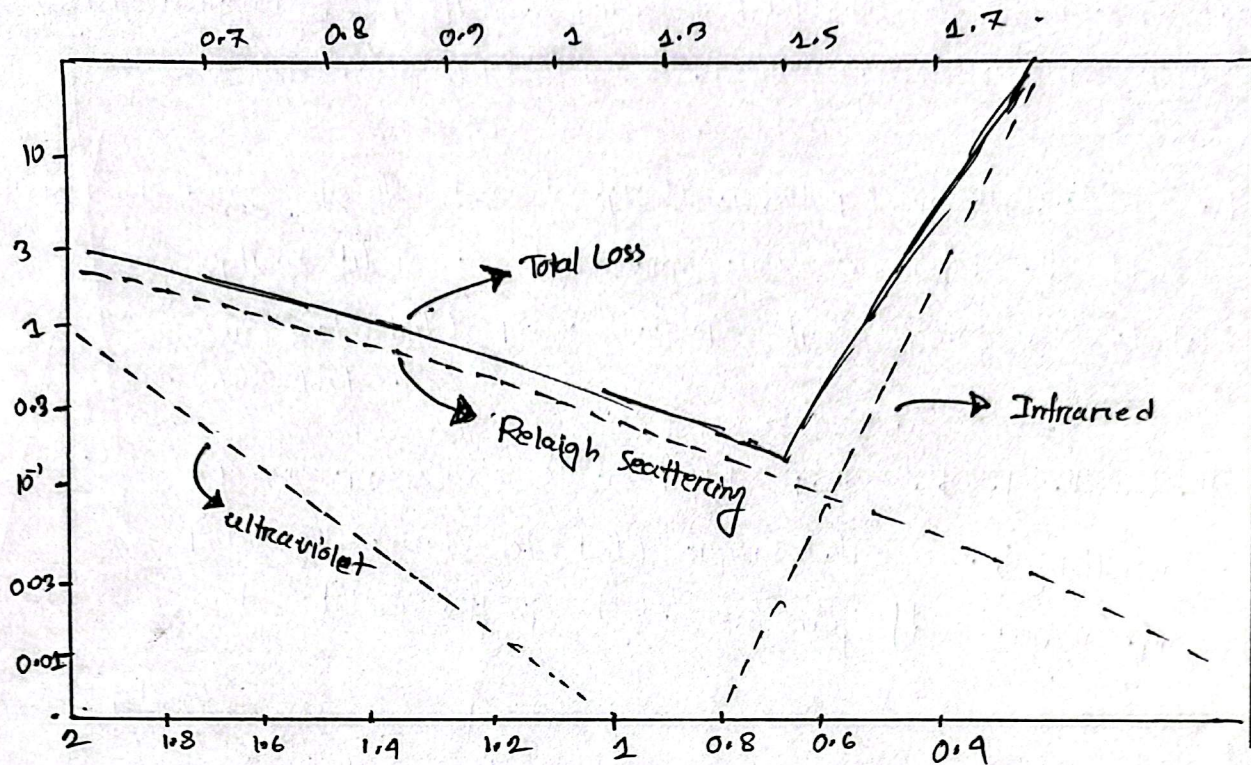
The intrinsic absorption loss mechanism with necessary illustration

Total loss curve: The total loss curve in the material is the sum of different attenuation shown by the top solid line in the graph. This includes loss due to Rayleigh scattering, ultraviolet and infrared absorption.

Rayleigh scattering: is the scattering of light due to microscopic variation in the refractive index, cause signal attenuation

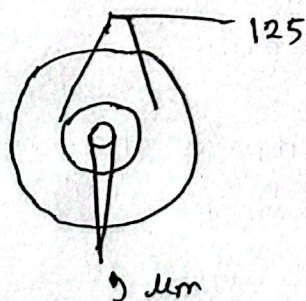
Ultraviolet absorption: The dash line indicates UV absorption, which increase at shortest wavelength, and it occurs due to material absorption.

Infrared absorption: At longer wavelength infrared absorption becomes dominant. This due to vibrational mode in the glass structure, that absorb longer wavelength.



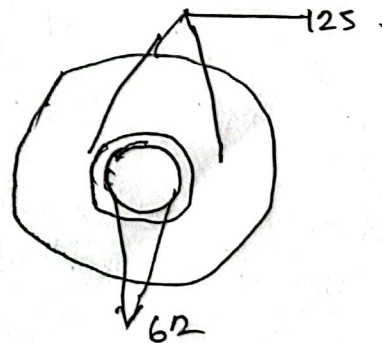
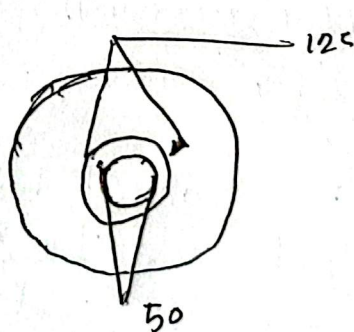
Single mode fiber cable

- A fiber optic cable with small core that allow only one mode of light to pass
- core $\rightarrow$ 9 microns, cladding - 125
- for long distance communication
- Low data capacity and high signal quality over long distance



Multimode fiber cable

- It fiber optic cable with large core that allow multiple mode of light to pass.
- core - 50/62, cladding - 125
- for short distance communication
- High data capacity but quality reduce over long distance



Passive Component

Passive Component: In optical network, passive component are devices that do not require electrical power to operate signal. They function using the physical properties of light (reflection, refraction, interference and absorption).

Types of physical component:

- MUX, DEMUX
- optical switch
- wavelength converter
- Couplers
- circulator and Isolator
- Attenuation
- optical filter

Exercise : 1

Current power = +3 dBm

Target power = -5 dBm

Required Reduction: $P_{\text{power in}} - P_{\text{power max}}$

$$= 3 - (-5)$$

$$= 8 \text{ dBm}$$

An attenuation of at least 8 dBm is required.

Exercise : 2

Since it is a 1:4 symmetric split: $P_{\text{out}} = \frac{10 \text{ mW}}{4} \approx 2.5 \text{ mW}$

$$\therefore \text{output power} \cdot P(\text{dBm}) = 10 \log_{10} P_{\text{mW}}$$

$$= 10 \log_{10} (2.5)$$

$$= 3.98 \text{ dBm}$$

Physical limitation of signal conversion

Why this matter for ESE students:

⇒ Understanding the physical limitation of signal conversion is crucial for power budgeting. If the light signal becomes too weak due to fiber loss, the photocurrent will drop below the noise floor, leading to a high Bit error rate (BER)

In optical communication system, the receiver's job is to convert light back into electricity.

- (i) Quantum efficiency (η): success rate of photodiode
- (ii) Responsibility (R): Responsibility tells us exactly how many amperes of current we get for every watt of light power.
- (iii) photocurrent (I_p): This is the actual electrical signal that will be amplified and process by the rest of the computer system to recover the transmitted data

Exercise : 1

$$\begin{aligned}\text{Responsibility} \cdot R &= \frac{\eta q}{h\nu} \\ &= \frac{\eta q \lambda}{hc} \\ &= \frac{0.7 \times 1.6 \times 10^{-19} \times 1310 \times 10^{-9}}{6.626 \times 10^{-34} \times 3 \times 10^{10}} \\ &= 0.739 \text{ A/W}\end{aligned}$$

Exercise : 2

$$\begin{aligned}\text{Photocurrent } I_p &= R \times P_{in} \\ &= 0.739 \times 5 \times 10^{-6} \\ &= 3.695 \times 10^{-6} \text{ A} \\ &= 3.695 \mu\text{A}\end{aligned}$$

Requirement of transmitter and receiver in WDM network

WDM: In WDM network, multiple data signals are sent simultaneously over a single optical fiber by using different wavelength of laser light. To make this work, the hardware - transmitters, receivers, and amplifiers - must meet specific high performance requirement.

Transmitter Requirement:

- Precision wavelength
- stability
- Narrow bandwidth
- wavelength locker.

Receiver Requirement

- Selectivity
- sensitivity
- wide bandwidth

Types of tunable laser

External cavity laser: Use a physical grating that moves to change the color

Distributed Bragg reflection: Use electrical current to change the refractive index of a built in grating, changing the wavelength.

Vertical cavity surface emitting laser: Small and cheap. They tune by moving a tiny top mirror using MEMS technology.

MEMS Technology: Combination of electrical and mechanical technology.

Optical amplifiers:

Semiconductor amplifier: Uses semiconductor chip

Erbium doped fiber amplifier: A piece of glass fiber is doped with erbium ions.

Erbium Bismuth fiber amplifier: Same to EDFA but uses Bismuth.

Exercise: 1

$$\begin{aligned} \text{(i) Output power} &= P_{in} + P_{gain} \\ &= -20 + 25 \\ &= 5 \text{ dBm} \end{aligned}$$

$$\begin{aligned} \text{(ii) first convert 5 dBm into mW} \\ &= 10^{\left(\frac{5 \text{ dBm}}{10}\right)} = 3.16 \text{ mW} \end{aligned}$$

$$\text{Total power} = 8 \times 3.16 = 25.28 \text{ mW}$$

Signal Dispersion

Material Dispersion: This occurs because the refractive index of glass varies with the wavelength of light.

Since the LED light source isn't pure, different colors travel at different speeds through the fiber.

Pulse Broadening: Parts of light pulse arrive at different times due to speed. The pulse spreads out or broadens as it travels.

Intersymbol Interference: For CSE student, the practical impact is ISI. If pulse broadens too much, it starts overlapping with the pulse representing the next bit.

Exercise: 1

Pulse Broadening due to

material dispersion $\Delta T = D_{mat} \times L \times \Delta \lambda$

$$= 80 \times 20 \times 90$$

$$= 6400 \text{ ps}$$

$$= 6.4 \text{ ns}$$

$$D_{mat} = 80 \text{ ps/(nm-km)}$$

$$L = 2 \text{ km}$$

$$\Delta \lambda = 90 \text{ nm}$$

$$\text{Maximum Bit Rate } (R_B) = \frac{1}{2\Delta T} = \frac{1}{2 \times 6.4 \times 10^{-9} \text{ s}}$$

$$= 78,125,000 \text{ bps}$$

$$= 78.13 \text{ Mbps}$$

Optical feedback and laser oscillation

Mechanism of optical feedback: When a light from a laser reflect back into laser, it is called optical feedback.

Laser: A laser consist of an active medium, placed between two mirrors forming an optical cavity.

As the light pulse through the medium repeatedly, it triggers stimulated emission, which amplifies the signal.

oscillation occurs when the gain within the medium exactly matches the loss.

Why it is a problem:

- Makes the laser unstable
- Cause power fluctuation
- Distort the signal

Laser oscillation:

- light bounces back and forth inside the cavity
- Each time its get amplified.
- Finally a strong beam comes out.

Fusion splicing: A method that permanently join two optical fibers.

how it works:

- fibers are heated
- Melted ends are fused together

Advantage:

- Low signal loss
- High Reliability
- No Reflection

Disadvantage

- High cost
- precision required
- fragility.

Example 8.1

input power (P_1) = 60 μ W

output power (P_2) = 0.004 μ W

output power (P_3) = 26 μ W

output power (P_4) = 27.5 μ W

Excess Loss: $10 \log_{10} \left(\frac{P_1}{P_3 + P_4} \right)$
 $= 10 \log_{10} \left(\frac{60}{26 + 27.5} \right)$
 $= \boxed{0.5 \text{ dB}}$

Insertion loss	Cross talk	Split ratio
Port 3: $10 \log_{10} P_1/P_3$ $= 10 \log_{10} 60/26$ $= \boxed{3.63 \text{ dB}}$	$10 \log_{10} (P_2/P_1)$ $= 10 \log_{10} (0.004/60)$ $= \boxed{-41.76 \text{ dB}}$	Port 3: $(P_3 / P_3 + P_4)$ $= \boxed{48.6 \%}$
Port 4: $10 \log_{10} P_1/P_4$ $= 10 \log_{10} (60/27.5)$ $= \boxed{3.39 \text{ dB}}$		Port 4: $(P_4 / P_3 + P_4)$ $= \boxed{51.4 \%}$

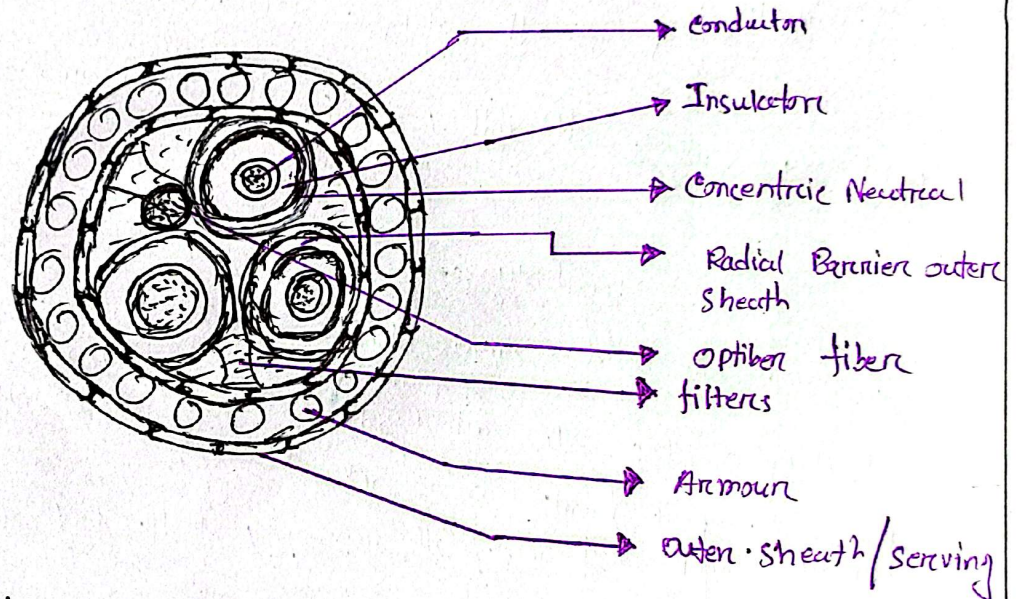
Optical Fiber cable design

Types of Buffering:

Tight Buffering: A thick layer of plastic is applied directly onto the fiber coating.

Loose tube Buffering: A fiber sits loosely inside a larger, hard plastic tube which is often filled with a water resistant gel.

- This allows the fiber to move slightly when the cable expands or contracts due to temperature.



Central strength member: A Rod made of steel or fiber glass placed in the center of the cable to provide rigidity.

Aramid yarns: Strong synthetic fibers around the buffer tubes to provide high tensile strength allowing the cable to be pulled through conduits without breaking.

Fabry perot laser

- This laser uses two mirrors at the end of the semiconductor material.
- Multiple light with different wavelengths.
- More dispersion.
- EX: Flash light.

Distributed feedback laser

- DFB laser has a Bragg grating etched directly into the semiconductor.
- Single wavelength.
- Less dispersion.
- Laser light.

Fabrication of the Fiber Couplers: Fiber couplers are passive component used to split or combine light signal.

Fused Biconical Taper Method: Two or more fiber stripped of their coating, placed side by side, and twisted together.

Polished Fiber Method: Two fiber are parallel partially embedded in separate glass block:

